Year 8 Mathematics

Unit 1: Number

1.4 Priority of Operations

Priority of Operations

STARTER · 5 MIN

Date: _____ **Title: Priority of Operations**

LEARNING OBJECTIVES

- To be able to follow the priority of operations

STARTER *Answer in your book. No calculators.*

1. Work out $5 + -12$
2. Work out $5 - -12$
3. Work out -8×-7
4. Work out $(-7)^2$

Starter — Answers

STARTER

Tick or correct in green pen

1. **−7**

2. **17**

3. **56**

4. **49**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can recognise and recall the order of operations using BIDMAS	☐
SILVER	I can apply BIDMAS correctly to solve calculations with more than one operation	☐
GOLD	I can solve complex problems using brackets, indices and mixed operations with clear reasoning	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Priority of Operations

INTRODUCTION · 3 MIN

BIDMAS gives the order to work out a calculation with more than one operation.

B

Brackets first.

$(7 - 5)^4 = 2^4 = 16$

I

Indices: powers and roots.

$3 + 2^2 = 3 + 4 = 7$

D M

Divide and multiply, left to right.

$6 + 4 \times 3 = 6 + 12 = 18$

A S

Add and subtract, left to right.

$18 - 12 \div 3 = 18 - 4 = 14$

Examples 1 and 2 — Using BIDMAS

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint B I D M A S × and ÷ before + and −

Example 1

Work out

$6 + 4 \times 3$

Example 2

Work out

$18 - 12 \div 3$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Multiply before you add

$6 + 4 \times 3 \quad = 6 + 12$

$\quad\quad\quad\quad\quad\quad = \mathbf{18}$

Tip: not $10 \times 3 = 30$.

Example 2

Divide before you subtract

$18 - 12 \div 3 \quad = 18 - 4$

$\quad\quad\quad\quad\quad\quad = \mathbf{14}$

Tip: D and M come before A and S.

Task 1

PRACTICE · 9 MIN

BRONZE *Work out*

Hint B I D M A S × and ÷ before + and −

1. $3 + 5 \times 2$

2. $20 - 6 \times 3$

3. $7 + 18 \div 6$

4. $4 \times 5 - 3 \times 2$

5. $30 \div 5 + 2 \times 4$

★ $10 - 8 \div 2 + 1$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 **13**

2 **2**

3 **10**

4 **14**

5 **14**

★ **7**

Examples 3 and 4 — Brackets and Indices

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Brackets first Then powers

Example 3

Work out

$100 \div (5 \times 2) + 3$

Example 4

Work out

$2 \times (5 - 1)^3$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Brackets first

$100 \div (5 \times 2) + 3$

$= 100 \div 10 + 3$

$= 10 + 3$

$\mathbf{= 13}$

Tip: then divide, then add.

Example 4

Brackets, then the power

$2 \times (5 - 1)^3$ $= 2 \times 4^3$

$= 2 \times 64$

$\mathbf{= 128}$

Tip: $4^3 = 4 \times 4 \times 4$.

Task 2

PRACTICE · 9 MIN

SILVER *Work out*

Hint Brackets first Then powers

1. $(7 - 5)^4$

2. $7 \times 5 - 1^4$

3. $3 \times (2 + 4)$

4. $(12 - 3) \div 3$

5. $5 + 3^2$

★ $(2 + 3)^2 - 4$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. 16
2. 34
3. 18
4. 3
5. 14
★ 21

Examples 5 and 6 — Negatives and Reasoning

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $(-3)^2 = 9$ Test brackets with BIDMAS

Example 5

(2 marks)

Work out

$(18 \div 3 + 2) \times -4$

Example 6

(1 mark)

Add brackets to make it correct.

$5 + 3 \times 4 - 2 = 30$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Inside the brackets: ÷ first

$18 \div 3 + 2 \quad = 6 + 2$

$\quad = 8$

$8 \times -4 \quad = -32$

Tip: different signs: negative.

Example 6

Try brackets in different places

$(5 + 3) \times 4 - 2 \quad = 8 \times 4 - 2$

$\quad = 32 - 2$

$= 30$

Tip: test each answer with BIDMAS.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: BIDMAS*

Hint $(-3)^2 = 9$ Test brackets with BIDMAS

1. $-6 + 12 \times -2$

2. $40 \div -5 + 18$

3. $(2 \times 5)^5 - 4^2$

4. $-100 \div (-20) + 15$

5. Add brackets:
   $8 - 3 \times 2 = 10$

★ Add brackets:
$2 + 6 \div 2 \times 4 = 1$

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. **−30**

2. **10**

3. **99 984**

4. **20**

5. **(8 − 3) × 2**

★ **(2 + 6) ÷ (2 × 4)**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE True or false: $6 + 4 \times 3 = 30$

SILVER True or false: $18 - 12 \div 3 = 14$

GOLD True or false: $100 \div (5 \times 2) + 3 = 13$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **False: 6 + 12 = 18**

SILVER **True: 18 − 4**

GOLD **True: 10 + 3**

Exam Practice — Order of Operations

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint B I D M A S × and ÷ before + and −

Question 1

(2 marks)

Work out

(a) 9 + 6 × 2 (1)

.....

(b) 24 − 15 ÷ 3 (1)

.....

Question 2

(2 marks)

Kim says 4 + 6 × 5 = 50.

Is she correct? Explain.

.....

Exam Practice — Brackets and Powers

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Brackets first Then powers

Question 3

(2 marks)

Work out $3 \times (8 - 2)^2$

.....

Question 4

(2 marks)

Work out $(4 + 2)^2 \div 9 + 1$

.....

Exam Practice — Reasoning with BIDMAS

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $(-3)^2 = 9$ Test brackets with BIDMAS

Question 5

(2 marks)

Work out $(-3)^2 - 4 \times -5$

.....

Question 6

(2 marks)

Put brackets in each calculation to make it correct.

(a) $3 + 4 \times 5 = 35$ (1)

.....

(b) $20 - 8 \div 4 + 2 = 2$ (1)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 21 B1

(b) 19 B1

Tip: × and ÷ first.

Question 2

No: multiply first B1

4 + 30 = 34 B1

Tip: she added first.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$3 \times 6^2 \quad = 3 \times 36$ **M1**

$\mathbf{= 108}$ **A1**

Tip: brackets, then the power.

Question 4

$6^2 \div 9 + 1 \quad = 36 \div 9 + 1$ **M1**

$\mathbf{= 5}$ **A1**

Tip: 36 ÷ 9 = 4.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$9 - -20 = 9 + 20$ M1

$= 29$ A1

Tip: $(-3)^2 = 9$.

Question 6

(a) $(3 + 4) \times 5$ B1

(b) $(20 - 8) \div (4 + 2)$ B1

Tip: check each with BIDMAS.